Operators in java

* Type promotion Rules
→ byte short & char values are promoted to int
→ If one operand is long, the whole expression will become long.
→ If one operand is float, entire expression will become float
→ If one operand is double, entire expression is double

* Operators
* 1) Arithmetic operators
:- (+, −, *, /, %.)
(+=, −=, *=, /=, %=,
++, −−)

* Relational Operators
→ ==, !=, <, >, <=, >=
→ used in else if conditions

```
int a = 5;
int b = 10;
a == b; // false
a = b  —→ a is assigned the value of b
```

* Bitwise Operators
→ Bit Manipulation
→ &, |, ^, ~, >>, <<

	&	and
~~and~~	|	bitwise and
	^	bitwise exclusive or
	~	~~bitwise~~ not (unary operator)
	>>	shift right
	<<	shift left
	>>>	shift right with zero will , & so on....

(>>=, <<=, >>>=, |=, &=, ^=)

byte ~~int~~ a = 2;
byte ~~int~~ b = 3;
int c = a&b; → Integer

2 ⟶ 10
3 ⟶ 11

a = 0 0 0 0 0 0 1 0
b = 0 0 0 0 0 0 1 1 &

a	b	a&b	a|b	a^b	~a	~b
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	1
1	1	1	1	0	0	0

Truth Table

c = c&2; ⇒ c&=2;

Shift operators & same for right shift just shifting is done in right
left shift (<<)

byte b = 8; 0 0 0 0 1 0 0 0 = 8
b = b<<1; 0 0 0 1 0 0 0 0 = 16 ×2
b = b<<1 0 0 1 0 0 0 0 0 = 32 ×2

b<<1; ⇒ b = b*2';
b<<2: ⇒ b = b*2²

by going on b becomes zero at one time & becomes constant

<< or >> → (perform only on)
- ↳ int, long ✓ ←
- ↳ byte short ×

(if doing byte, short they get promoted to int)

int i = 1
then i << 31 ; Max^n MSB
after that it does i<<(32%32)
i.e mod of 32

Right shift (>>) : opp of left shift

>>> shift right with zeros
↳ dumb shift

byte b = -128; 1 0 0 0 0 0 0 0 = -126
b>>1; 1 1 0 0 0 0 0 0 = -64
b>>>1; 0 1 0 0 0 0 0 0 = +64

* Logical Operators
→ && (AND) this doesn't work on bits but on expressions
|| (OR)
! (NOT)

int a = 5;
int b = 10;
int c = 15;

		AND	OR
True	false	False	True
false	True	False	True
True	True	True	~~false~~ True
false	false	False	false

boolean and = (a < b) && (a<c)
and → True; (a < b) → T; (a<c) → T

Short Circuit

A	B	A && B
F	F	F
F	T	F
T	F	F
T	T	T

A && B
↓
F

If any one is false then don't check B because AND will give false

& this concept is called Short circuit

& opposite goes for OR operation

#Bitwise operators

```
int a = 5
    b = 10
    c = 6;
d = (a<b) & (a<c)
```

we can use bitwise operators here as well

but it does not do short circuiting

* Assignment operator

```
int a = 5;
```

int b b = a; → value of a is assigned to b

* Ternary operator
→ used in conditionals

* operator precedence

```
int c = a*b + d - f/g << 2;
```

like in Maths we've BODMAS
in java we have precedence of operators which also follows BODMAS

lowest precendence is of assignment (=) operator;
highest precedence is of bracket ();